J Syst Sci Complex (2026) 39(4): 1436–1462

Stochastic Hybrid Rumor Control: A Data-Driven Ensemble Learning Control Algorithm*

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DOI: 10.1007/s11424-025-4372-4
Received: 31 July 2024 / Revised: 30 April 2025
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Abstract To explore the impact of various groups and methods on rumor propagation, the authors propose a 'Double-Refutation (DR) and Double-Blocking (DB)' rumor control strategy. This strategy combines external refutation via media reports, internal refutation by counteracting individuals, and both continuous and impulse blocking methods. By leveraging multi-synergy and aiming to minimize control costs, the authors propose stochastic optimal hybrid control strategies for rumor containment. Additionally, to enhance the response speed of the control strategy, the authors introduce an ensemble learning algorithm as a substitute for theoretical solutions. Numerical simulations demonstrate that the trained ensemble learning control algorithm can quickly identify sub-optimal control strategies for rumor spreading, with costs only 4.1% higher than those of the optimal control theory.

Keywords Double refutation and blocking control mechanism, dual-layer propagation model, hybrid optimal control, stacking ensemble learning control algorithm, stochastic control.

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*This research was supported by the National Natural Science Foundation of China under Grant No. 62403153.
◇*This paper was recommended for publication by Editor TAN Ying.*

1 Introduction

With the advancement of the process of information industrial revolution, online social networks have entered the rapid development of popularity, and have gradually become an important platform for various media and groups to disseminate information. However, false information such as rumors have sprung up on social platforms[1]. These false news distort the original facts and maliciously slander different groups, which may cause innocent people to suffer, the government loses credibility, social panic and huge economic losses[2] in a certain period of time.

Due to negative influence of rumors, people's research on the process of rumor propagation has been continuing up to now. Among them, it has become a crucial research direction to establish appropriate models and design corresponding control strategies based on the process of rumor propagation[3–9]. From the perspective of the intervention methods used, the researches on rumor control can be divided into the following two categories: One is to counter and clarify rumors by releasing truth, official news and other real information[10–14]. For example, Yin, et al.[10] proposed a model based on the amount of public opinion to understand how to suppress the spread of rumor information and strengthen the spread of anti-rumor. Xiao, et al.[11] proposed a rumor and anti-rumor propagation model based on data enhancement and evolutionary game, which improves the prediction accuracy of users' behaviors in information propagation. Dong, et al.[12] established a two-layer rumor control model considering the control strategy of delay and event trigger impulse, and used the media layer subject in the model to release truth to debunk rumors. Xiang, et al.[13] proposed a rumor heat prediction model based on multiple rumors, anti-rumor information and knowledge representation, which effectively predicted the group behavior and heat of the rumor topic and revealed the mechanism behind the rumor topic propagation; Mou, et al.[14] proposed an online rumor diffusion method based on rumor, anti-rumor and motivational rumor, and put it into a rumor diffusion model based on dynamic and evolutionary game theory to effectively predict group behavior under rumors and reflect the cooperation and competition among multiple information; The second is to block or forcibly isolate major groups and media that play a significant role in rumor spread, so as to cut off the spread path of rumors[15–19]. For example, Ding, et al.[15] designed a discrete control strategy that uses impulse control to isolate rumor spreaders, and obtained the optimal control strategy of their rumor control model through calculus of variations. Yu, et al.[16] proposed a heterogeneous rumor spreading model with two language spreaders, and considered the impulsive isolation control strategy. Wang, et al.[17] proposed a dynamic rumor minimization user experience impact model, which minimizes the impact of rumors by shielding a specific subset of nodes. Yang, et al.[18] proposed a mathematical programming formula based on integer linear programming to minimize rumor spread by blocking a subset of nodes in a complex social network. Ni, et al.[19] proposed a community-based rumor blocking problem, which selects a group of torrent users as protectors to maximize the expected number of users who are ultimately not affected by the rumor source. In terms of the effect of intervention measures, among the known research results, Wen's study[20] proved that implementing multiple intervention

measures at the same time has better control effect than considering only single intervention measures. From the perspective of control theory, hybrid control strategies have been applied widely in the area of control[21]. However, the hybrid control strategies of individual and media rumor-refuting behavior, continuous and discrete blocking behavior, have not been discussed in detail. Additionally, it is also challenging to solve the optimal control strategy under multiple complex factors.

The implementation of control strategies invariably incurs substantial economic implications[15], necessitating a meticulous assessment of the associated costs and potential benefits. For example, when rumors are generated, administrators of network media platforms need to intensify efforts to completely block relevant information, close relevant accounts[22], delete relevant posts[23] and monitor and supervise the relevant content[24] to prevent the spread of rumors. To some extent, these measures violate the original intention of opening up online media platforms, and infringe on the freedom of speech of some netizens. Furthermore, the guiding role of the official media in the spread of rumors should not be underestimated[25–29]. Generally, ordinary netizens are easily affected by truth verified by the official media, so as to play the guiding role of the refuters in the process of information transmission to counter rumors, thus reducing the influence scope of rumors and controlling the cost of rumor transmission. The trade-off between controlling rumor spread and controlling cost raises an optimal control problem. The optimal control problem in rumor spread has been discussed by many scholars have discussed the optimal control problem in rumor spread[12, 15, 30–39]. For spreading models described by ordinary differential equations, researchers[30−35] derived the necessary conditions for optimal control using variational methods and Pontryagin's minimum principle. For stochastic differential equation models[36–39], scholars applied the stochastic maximum principle, reconstructed cost functions and Hamiltonians, and derived stochastic optimal solutions. Ding, et al.[15] analyzed necessary conditions for hybrid optimal control for models with impulses. Notably, there remains a lack of analysis and verification for optimal control problems in hybrid stochastic models with impulses. Summarizing the above analysis and research, we extracted four key factors model structure, model layers, stochastic environment, and control strategies to conduct a comparative analysis of existing literature, as shown in Table 1.

Table 1 Comparisons of existing models about rumor propagation and control

Paper	Model name	Model structure	Stochastic model	Control strategies
Yin, et al.[10]	$S_1S_2I_{S1}I_{S2}$ $R_1R_2E_1E_2D$	Single Layer	N	Refutation: Anti-rumors information in propagation process
Dong, et al.[12]	$XYZISR$	Dual Layers	N	Refutation: Debunking behavior
Yu, et al.[16]	$S^1S^2I^1I^2RQ$	Single Layer	N	Blocking: Isolation major nodes

Table 1 (Continued) Comparisons of existing models about rumor propagation and control

Paper	Model name	Model structure	Stochastic model	Control strategies
Sun, et al.[17]	$SEIHR$	Single Layer	N	Blocking: Three synergistic control strategies of prevention, correction and quarantine
Zhong, et al.[5]	$SIRHD$	Single Layer	Y	Refutation: Anti-rumors information in propagation process
Zhang, et al.[4]	$S_N S_B I_B R$	Dual Layers	N	Blocking: Using public credibility to block rumors
Jain, et al.[9]	$S_1 S_2 I$	Single Layer	N	Blocking: Experts and governments implement interventions
Ding, et al.[15]	$SI_1 I_2 RQ$	Single Layer	N	Refutation: Anti-rumor spreaders in propagation process Blocking: Temporary isolation of individuals
Wang, et al.[22]	$SMIRT$	Single Layer	N	Blocking: Reducing rumor propagators
Guo, et al.[29]	$SEIMR$	Single Layer	N	Refutation: Media propagating anti-rumors information
Ding, et al.[31]	$SI_1 I_2$	Single Layer	N	Blocking: Reducing rumor propagators
This paper	$SI_1 I_2 B$ MXY	Dual Layers	Y	Refutation: Spread clarification messages Blocking: Isolate major individuals

Based on the above considerations, a DR and DB hybrid control strategy is designed in this paper, and a dual-layer rumor control model is established. After rumors begin to spread, the government, the refuting individual and the media begin to implement control measures. The government can directly block the spread of rumors, and individuals can resist the spread of rumors by releasing truth. The media is also capable of spreading information and guiding public opinion, and plays an active role in the process of rumor spread. By refuting and blocking the rumors, the status of the people related to the rumors was changed and the goal of controlling the spread was achieved. The whole process was shown in Figure 1. Generally, in the first stage, rumor spreaders will spread rumors to influence rumor no-contact people, causing them to change their status to rumor believers and rumor spreaders. When rumors appear, rumor refuting media and the government will adopt direct and indirect control strategies to stop the spread of rumors. After that, rumor spreaders, rumor no-contact people and rumor believers will have the possibility to turn into rumor refuters and removers. Meanwhile, rumor refuters influence the rest of the population by releasing clarification messages, and finally bring the rumor under control. Moreover, we introduce an ensemble learning algorithm to make up

for the shortcomings of optimal control method, so as to improve the response speed of control strategy and improve the hysteresis defect of control strategy. As shown in Figure 2, the main contributions of this paper are summarized as follows:

1) Dual-layer hybrid rumor control model: A dual-layer network model is established to accurately describe the whole process of rumors from spreading to being controlled and eliminated. Two rumor control strategies are applied in the model which are DR control strategies (individual-induced refutation and media-induced refutation) and DB control strategies (continuous blocking method and impulse blocking method).

2) Optimal control ensemble learning control algorithm: An optimization problem is formulated and the optimal control method is applied to solve the problem of minimizing the control cost. In order to enhance the adaptability of the optimal control method, the integrated learning algorithm is innovatively introduced into the optimal control problem to quickly solve the optimal control strategy.

3) Data simulation experiment: Numerical simulation experiments are designed to verify the effectiveness of the dual-layer coupled dynamics model of rumor propagation and control model, and the experimental results show that the ensemble learning algorithm can quickly find the sub-optimal control strategy for rumor spread, and its cost is similar to the optimal control theory.

The structure of this paper is as follows: In Section 2, the dual-layer rumor control model mentioned above is established, and the role of the DR and DB hybrid control strategy in the model is described in detail. In Section 3, the problem of cost minimization is proposed, the optimal control theory is used to deduce and solve the problem, and the optimal solution is obtained. In Section 4, numerical simulation experiments of optimal control theory and ensemble learning algorithm are carried out, and the cost under six other different cases is compared with the cost under optimal control to further verify the cost minimization of the dual-layer rumor control model. In Section 5, we draw a conclusion.

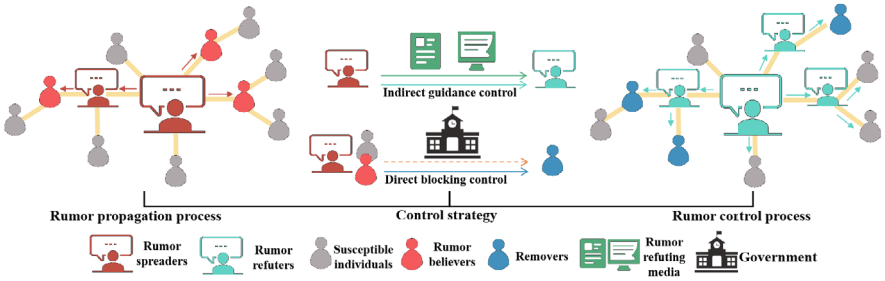


Figure 1 Diagram of rumor propagation and control. Rumor propagation process: Rumor spreaders start to propagate rumors and quickly affect Susceptible individuals, the thick lines between in figures are the social connection relationship, and the arrows are the propagation relationship. Control strategy includes direct blocking by government departments and indirect guidance by the media and Rumor refuters. Among them, indirect guidance leads Rumor spreaders to Rumor refuters and starts to spread the truth. Rumor control process: Includes the group temporarily banned from speaking (marked with blue human shapes), and Rumor refuters spreading the truth

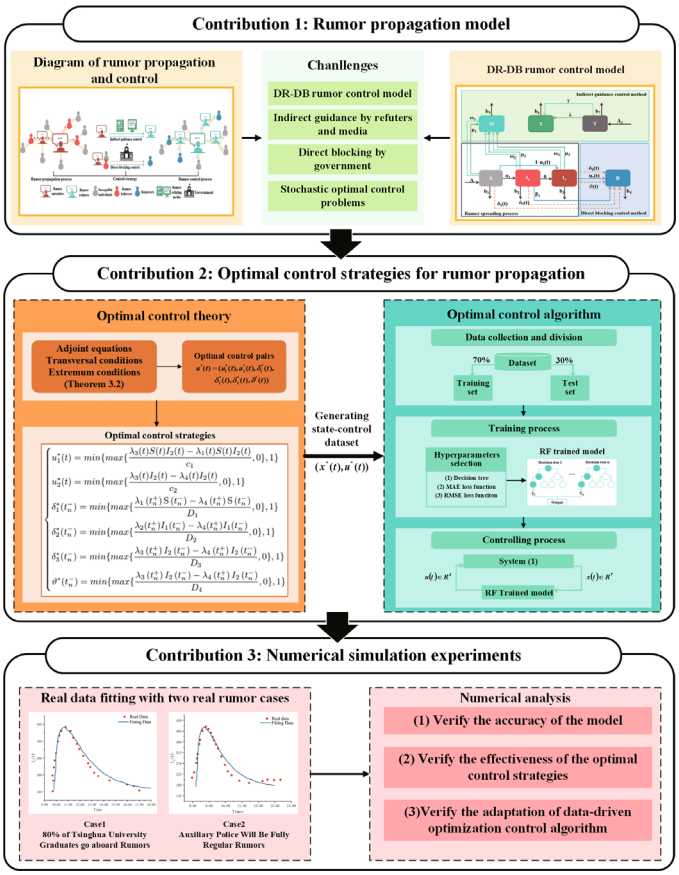


Figure 2 Framework and contributions of the paper

2 The DR-DB Rumor Control Model

This section develops a $DR-DB$ control strategy to mitigate the negative impacts caused by rumor propagation. To tackle this control process, it is imperative to first model the issue. Accordingly, this section unfolds in three steps: 1) Introducing the population-initiated model, 2) formulating a $DR-DB$ strategy, and 3) constructing the $DR-DB$ control model framework.

2.1 Population-Initiated Model

Based on the population's receptivity to rumors, individuals are categorized into three distinct groups: Susceptible (S), individuals who have not yet encountered the rumor; Believers-Non-Spreaders (I_1), those who believe the rumor but do not disseminate it; and Believers-Spreaders (I_2), individuals actively spreading the rumor.

Within a unit time t, the probability of netizens engaging in rumor-related topics is denoted by A, while the probabilities of the aforementioned three groups losing interest in the rumor are represented by b_1, b_2 and b_3, respectively. Moreover, the probability of individuals in group S being influenced by those in group I_2 and transitioning to group I_1 is denoted by α_1, and the probability of moving directly from group S to I_2 is α_2. Additionally, the probability of

individuals in group I_1 being influenced by group I_2 and converting to I_2 is represented by θ.

The transitions in states during the rumor control process are depicted in Figure 3. We represent the original rumor spread process with black line rectangle, where we use rounded-rectangles to represent the population S, I_1 and I_2, and the black arrow indicates the route of rumor spread.

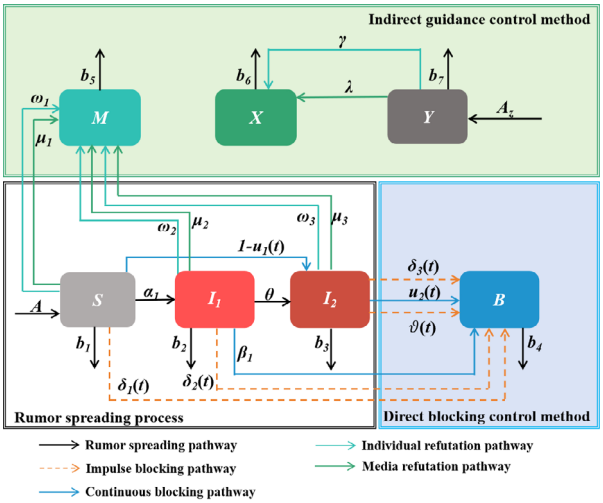


Figure 3 Flow of the dual-layer rumor spreading control model with DR and DB hybrid control strategy. The top rectangle and the bottom right rectangle represent respectively DR and DB control strategies

2.2 DR-DB Control Strategy

On the basis of the original rumor spreading process, we add the control strategy of DR and DB. In Figure 3, we aligned it into indirect guidance control method and direct blocking control method, represented by the rectangle of light green background and the rectangle of light blue background respectively.

2.2.1 DR Control Mechanisms

This subsection is divided into two parts, focusing on the mechanisms of refuting rumors through dual channels: Spontaneously by the population and through media intervention.

Population-initiated refutation: Refutation efforts originating within the population, such as opinion leaders, classified as Rumor Refuters (M), which implies individuals who do not believe in the rumor and disseminate clarification messages. These individuals influence the state of S, I_1, I_2 groups, aiming to control the I_1 and I_2 populations through the dissemination of clarifying messages. Within a unit time t, the probability of M individuals losing interest in the rumor is denoted by b_5. The probabilities of S, I_1, and I_2 groups being influenced by M group's clarifying messages and transitioning to M group are represented by ω_1, ω_2 and ω_3, respectively. Considering the randomness in clarifying rumor processes, we assume the spread parameter ω_3 is affected by white noise $\dot{W}_1(t)$, formulating the population-initiated rumor refutation mechanism.

$$\omega_3 \to \omega_3 + \sigma_1 \dot{W}_1(t).$$

Media-initiated refutation: The second type of refutation control method is initiated by the media. The spread of rumors is closely associated with social media, which we divide into Debunking Media (X) and Susceptible Media (Y) two distinct states. Debunking Media (X) refers to media platforms that spread true information, such as the workplaces or schools of rumor victims, while Susceptible Media (Y) refers to media that have not yet focused on or reported the rumor. Within a unit time t, the probability of Y beginning to pay attention to the rumor is Az, and the probabilities of X and Y media losing interest in the rumor are b_6 and b_7, respectively. The probability of Y being influenced by M group's clarifying messages and transitioning to X is denoted by λ, and the probability of being influenced by X's debunking messages and transitioning to X is γ. The probabilities of S, I_1, and I_2 groups being influenced by X's debunking messages and transitioning to M group are μ_1, μ_2 and μ_3, respectively.

In the top rectangle of Figure 3, we use rounded-rectangles represent the population M, media X and Y, here join the main cor responding to the premise of DR control strategy, arrows represent the individual refutation control pathway caused by population M and media X.

2.2.2 DB Control Mechanisms

Rumor blocking strategies represent a common and effective control approach, wherein relevant authorities isolate individuals as silenced members of the population through direct blocking, denoted as group Blocking individuals (B). Considering the control costs and potential human rights issues associated with blocking strategies, we design a hybrid blocking strategy that combines continuous and impulse elements.

Continuous blocking: The first is a continuous blocking strategy, where the probability of individuals in group I_1 being blocked is β_1. Given that individuals in group I_2 are of particular control interest, we design a time-varying control intensity for I_2, represented by control strengths $u_1(t)$ and $u_2(t)$. Here, $u_1(t)$ denotes the punitive intensity applied by authorities for rumor spreading by I_2; $u_2(t)$ signifies the isolation intensity. Furthermore, acknowledging the stochastic nature of isolation measures, we model control intensity $u_2(t)$ as a stochastic variable driven by white noise $\dot{W}_2(t)$, specifically.

$$u_2(t) \to u_2(t) + \sigma_2 \dot{W}_2(t).$$

Impulse blocking: The second strategy is an impulse blocking strategy. At each impulse moment t_n, groups S, I_1, and I_2 are isolated into group B with impulse strengths $\delta_1(t_n)$, $\delta_2(t_n)$, and $\delta_3(t_n) \in [0, 1]$, respectively. Additionally, an impulse strength $\vartheta(t_n) \in [0, 1]$ is applied to further control the transition of group I_2 to B, ensuring precise isolation measures.

In the bottom right rectangle of Figure 3, we use rounded-rectangle to represent the population B, the solid arrows indicate the continuous blocking control pathway, and the dotted arrows indicate the impulse blocking control pathway, where two pathways correspond to the DB control strategy we had mentioned before.

2.2.3 DR-DB Control Model

In this paper, we adopt the following notation conventions. Consider a complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ equipped with a filtration $\{\mathcal{F}_t\}_{t\geq 0}$ that satisfies the usual conditions:

The filtration is increasing and right-continuous, and $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets. The expectation operator with respect to the probability measure is denoted by $\mathbb{E}[\cdot]$. Furthermore, we define an N-dimensional Brownian motion on this probability space as $W(t) = (W_1(t), W_2(t))^{\mathrm{T}}$, where each component $W_i(t)$, $1 \leq i \leq 2$ constitutes a mutually independent scalar Brownian motion. Based on the state classifications, control methods, and transition probabilities outlined above, we describe the rumor spreading control process as the following differential equations.

$$\begin{cases} dS(t) = [A - \alpha_1 S(t)I_2(t) - (1-u_1(t))S(t)I_2(t) - \omega_1 S(t)M(t) - \mu_1 S(t)X(t) \\ \qquad - b_1 S(t)]dt, \\ dI_1(t) = [\alpha_1 S(t)I_2(t) - \theta I_1(t)I_2(t) - \beta_1 I_1(t) - \omega_2 I_1(t)M(t) - \mu_2 I_1(t)X(t) - b_2 I_1(t)]dt, \\ dI_2(t) = [(1-u_1(t))S(t)I_2(t) + \theta I_1(t)I_2(t) - u_2(t)I_2(t) - \omega_3 I_2(t)M(t) \\ \qquad - \mu_3 I_2(t)X(t) - b_3 I_2(t)]dt - \sigma_1 I_2(t)M(t)dW_1(t) - \sigma_2 I_2(t)dW_2(t), \\ dB(t) = [\beta_1 I_1(t) + u_2(t)I_2(t) - b_4 B(t)]dt + \sigma_2 I_2(t)dW_2(t), \\ dM(t) = [\omega_1 S(t)M(t) + \omega_2 I_1(t)M(t) + \omega_3 I_2(t)M(t) + \mu_1 S(t)X(t), \\ \qquad + \mu_2 I_1(t)X(t) + \mu_3 I_2(t)X(t) - b_5 M(t)]dt + \sigma_1 I_2(t)M(t)dW_1(t), \\ dX(t) = [\lambda X(t)Y(t) + \gamma M(t)Y(t) - b_6 X(t)]dt, \\ dY(t) = [A_z - \gamma M(t)Y(t) - \lambda X(t)Y(t) - b_7 Y(t)]dt, \end{cases} \tag{1}$$

for $t \neq t_n$,

$$\begin{cases} S(t_n^+) = [1 - \delta_1(t_n^-)]S(t_n^-), \\ I_1(t_n^+) = [1 - \delta_2(t_n^-)]I_1(t_n^-), \\ I_2(t_n^+) = [1 - \delta_3(t_n^-)]I_2(t_n^-) - \vartheta(t_n^-)I_2(t_n^-), \\ B(t_n^+) = B(t_n^-) + \delta_1(t_n^-)S(t_n^-) + \delta_2(t_n^-)I_1(t_n^-) + \delta_3(t_n^-)I_2(t_n^-) + \vartheta(t_n^-)I_2(t_n^-), \\ M(t_n^+) = M(t_n^-), \\ X(t_n^+) = X(t_n^-), \\ Y(t_n^+) = Y(t_n^-), \end{cases} \tag{2}$$

for $t = t_n$. Here, t_n $(n = 1, 2, \cdots, N-1)$ is the impulsive blocking time, t_n^-, t_n^+ denote the previous and next moment of the impulsive blocking time t_n, and N is a finite positive constant. The algorithm of the proposed model is described as Algorithm 1.

Remark 2.1 The system (1) models the dynamical process of information dissemination. The drift term, diffusion term, and impulse function of the system are polynomial functions, with a finite number of impulses and transition relationships among the terms. As a result, the system admits a bounded feasible domain[7], which in turn ensures the satisfaction of the linear growth condition. Additionally, the system satisfies the local Lipschitz condition. Consequently, these properties establish the global existence and uniqueness of its solution[40].

Algorithm 1 The algorithm to implement the DR-DB control strategy

Require: The state of each individual in the population

Initialize: Select initial parameter values for α_1, α_2, β_1, $u_1(t)$, $u_2(t)$, $\theta(t)$, $\delta_1(t)$, $\delta_2(t)$, $\delta_3(t)$, ω_1, ω_2, ω_3, λ, γ, μ_1, μ_2, μ_3
for each time unit t in $[t_0, t_f]$ **do**
  for each individual $i = 1$ to $N(t)$ **do**
    for each individual $j = 1$ to $N(t)$ **do**
      if i is in contact with j **then**
        Double-Refutation (DR) Control:
        if i is influenced by M (Rumor Refuters) **then**
          if i is in S, I_1, or I_2 **then**
            Change state of i to M with probabilities ω_1, ω_2, ω_3
          end if
          if i is exposed to X (Debunking Media) **then**
            Change state of i to M with probabilities μ_1, μ_2, μ_3
          end if
          if Y (Susceptible Media) is influenced **then**
            Change state of Y to X with probability λ or γ
          end if
          Double-Blocking (DB) Control:
          Continuous Blocking for I_1:
          if i is in I_1 **then**
            Apply blocking with probability β_1
          end if
          Time-Varying Control for I_2:
          Adjust $u_1(t)$ and $u_2(t)$ for I_2, where $u_2(t) = u_2(t)$ + white noise $W_2(t)$
          Impulse Blocking at moments N:
          if current time t is an impulse moment **then**
            Isolate S, I_1, and I_2 into B with $\delta_1(t)$, $\delta_2(t)$, $\delta_3(t)$
            Apply additional impulse $\vartheta(t)$ to I_2
          end if
        else
          State of i remains unchanged
        end if
      end if
    end for
  end for
end for
return The updated state of each individual

3 The Optimal Control Method Based on Ensemble Learning Algorithm

In this section, we study optimal control to provide a theoretical basis for indirect guidance and direct blocking of rumor spreading. We first derive the optimality conditions using the stochastic maximum principle and apply them to solve the proposed optimal hybrid control problem in the rumor spreading system.

3.1 Necessary Condition of Optimality

We analyze the following stochastic hybrid system where the state variables are influenced by $N-1$ impulsive actions which are performed at times $t = t_n$.

$$\begin{cases} dx(t) = f(x(t), u(t))dt + g(x(t))dW(t), & t \neq t_n, \\ x(t_n^+) = x(t_n^-) + l(x(t_n^-), \upsilon(t_n^-)), & t = t_n, \end{cases} \tag{3}$$

with initial value $x(t_0) = x_0 \in \mathbb{R}^n$, $x(t; t_0, x_0)$ denotes the solution to the system (3), which represents the state and is assumed to be right continuous, i.e., $x(t_n^+) = x(t_n)$. The variables $u(t)$ and $\upsilon(t)$ represent the continuous and impulsive control inputs, respectively, used to regulate the system dynamics. $f : \mathbb{R}^n \to \mathbb{R}^n$, $g : \mathbb{R}^n \to \mathbb{R}^{(n\times m)}$ and $l : \mathbb{R}^n \to \mathbb{R}^n$ are Borel measurable functions.

We consider the quadratic cost function

$$J = \mathbb{E}\left\{\phi(x(t_f)) + \sum_{n=0}^{N-1}\left[\int_{t_n^+}^{t_{n+1}^-} L(x(t), u(t))\right]dt + \sum_{n=1}^{N-1} K\left[x(t_n^-), \upsilon(t_n), t_n\right]\right\}, \tag{4}$$

where $\phi(x(t_f))$ represents the terminal cost at the fixed terminal time t_f. L denotes the running and the continuous control cost, and K is the impulsive control cost.

Before stating the necessary conditions for optimal control, we need to make the following standing hypothesis.

Assumption 3.1 Assume that

(i) f, g, l are locally Lipschitz in x and satisfy the linear growth condition.

(ii) The admissible control sets U and V are defined as:

$$U = \{u_i(t) | 0 \le u_i(t) \le u_{\max}, \forall u_i \in \mathcal{L}^2_{\mathcal{F}}(t_0, t_f; R^P), i = 1, 2, \cdots, P\},$$
$$V = \{\upsilon(t_n) = (\upsilon_1(t_n), \upsilon_2(t_n), \cdots, \upsilon_Q(t_n)) | 0 \le \upsilon_j(t_n) \le \upsilon_{\max}, \upsilon_j(t_n) \in R^Q, j = 1, 2, \cdots, Q\},$$

where, $u_i^{\max}$ and $\upsilon_j^{\max}$ are positive constants. $\upsilon(t_n)$ represents the control value applied at the specific time t_n.

(iii) The running cost $L(x(t), u(t))$ is jointly convex in (x, u), the terminal cost $\phi(x(t_f))$ is convex in x, the impulsive cost $K(x(t_n^-), \upsilon(t_n), t_n)$ is convex in $\upsilon(t_n)$, and the impulsive dynamics $l(x(t_n^-), \upsilon(t_n))$ is convex in $\upsilon(t_n)$. Additionally, L, ϕ, K, l are differentiable and satisfy the quadratic growth condition.

Remark 3.2 Assumption 3.1 is proposed for the general system (3) to ensure the existence and uniqueness of solutions as well as the optimality of the control problem. Condition (i) on

functions f, g and l guarantees the global existence and uniqueness of solutions. Condition (ii) on the control sets U and V ensures the feasibility of the control. Condition (iii) on the cost function ensures the existence of an optimal solution.

Our goal is to find the hybrid controls $u^*(t) = (u_1^*(t), u_2^*(t), \cdots, u_P^*(t))$ and $\upsilon^*(t_n) = (\upsilon_1^*(t_n), \upsilon_2^*(t_n), \cdots, \upsilon_Q^*(t_n))$ such that $J(u^*, \upsilon^*) \leq J(u, \upsilon)$, $\forall u(t) \in U, \upsilon(t_n) \in V$.

Under Assumption 3.1, the sufficient conditions ensuring the existence of an optimal solution to Equation (4) can be derived from [37, Theorem 3.5], which we have omitted here for the sake of conciseness. In order to use the stochastic maximum principle and Hamiltonian methods, we introduce the continuous and impulsive Hamiltonian functions as follows:

$$\begin{cases} H_c = L(x(t), u(t)) + \lambda(t)^{\mathrm{T}} f(x(t), u(t)) + q(t)^{\mathrm{T}} g(x(t)), \\ H_{d,n} = K(x(t_n^-), \upsilon(t_n), t_n) + {\lambda_n}^{\mathrm{T}} l(x(t_n^-), \upsilon(t_n^-)), \end{cases} \tag{5}$$

where the vectors $\lambda(t)$, $q(t)$ and λ_n are costates adjoining the deterministic continuous, stochastic continuous and impulsive dynamics, respectively. Under Assumption 3.1, H_c is convex with respect to u, and $H_{d,n}$ is convex with respect to $\upsilon(t_n)$, ensuring the validity of the Hamiltonian minimization.

By applying the stochastic maximum principle[41, 42], the backward stochastic differential equations governing the adjoint variable $\lambda(t)$ are

$$\begin{cases} d\lambda(t) = -\dfrac{\partial H_c(x^*, u^*, \lambda, q)}{\partial x} dt + q(x^*(t)) dW(t), \\ \lambda(t_n^+) - \lambda(t_n^-) = -\dfrac{\partial H_{d,n}(x^*, \upsilon^*, \lambda_n)}{\partial x^*(t_n^-)}, \end{cases} \tag{6}$$

with the terminal condition:

$$\lambda(t_f) = -\frac{\partial \phi(x^*(t_f))}{\partial x}. \tag{7}$$

The optimal control $u^*(t)$ and $\upsilon^*(t_n)$ are obtained through Hamiltonian minimization:

$$\begin{cases} u^*(t) = \arg\min\limits_{u \in U} H_c(x^*, u^*, \lambda, q), \\ \upsilon^*(t_n) = \arg\min\limits_{\upsilon(t_n) \in V} H_{d,n}(x^*, \upsilon^*(t_n), \lambda_n), \end{cases} \tag{8}$$

which satisfies the first-order conditions:

$$\begin{cases} \dfrac{\partial H_c(x^*, u^*, \lambda, q)}{\partial u} = 0, \\ \left(\dfrac{\partial H_{d,n}(x^*, \upsilon^*, \lambda_n)}{\partial \upsilon} \right)^{\mathrm{T}} (\upsilon(t_n) - \upsilon^*(t_n)) \geq 0, \end{cases} \tag{9}$$

where the control sets U and V are convex domains defined in Assumption 3.1.

By simultaneously solving the state equations, adjoint equations, and optimality conditions,

Equations (6)–(9) can be reformulated as follows:

$$\begin{cases} dx^*(t) = \dfrac{\partial H_c(x^*, u^*, \lambda, q)}{\partial \lambda} dt + g(x^*(t))dW(t), \\ dx^*(t_n^-) = \dfrac{\partial H_{d,n}(x^*, \upsilon^*, \lambda_n)}{\lambda(t_n^+) - \lambda(t_n^-)} d(t_n^-), \end{cases} \tag{10}$$

$$\begin{cases} d\lambda(t) = -\dfrac{\partial H_c(x^*, u^*, \lambda, q)}{\partial x} dt + q(x^*(t))dW(t), \\ \lambda(t_n^+) - \lambda(t_n^-) = -\dfrac{\partial H_{d,n}(x^*, \upsilon^*, \lambda_n)}{\partial x^*(t_n^-)}, \end{cases} \tag{11}$$

$$\lambda(t_f) = -\frac{\partial \phi(x^*(t_f))}{\partial x}, \tag{12}$$

$$\begin{cases} \dfrac{\partial H_c(x^*, u^*, \lambda, q)}{\partial u} = 0, \\ \left(\dfrac{\partial H_{d,n}(x^*, \upsilon^*, \lambda_n)}{\partial \upsilon} \right)^{\mathrm{T}} (\upsilon(t_n) - \upsilon^*(t_n)) \geq 0. \end{cases} \tag{13}$$

Here, $x^*(t)$ denotes the optimal trajectory of $x(t)$.

3.2 Theoretical Solution of Optimal Control Problem

For the system (1)–(2), we use Theorem 3.3 to solve the specific optimal control solution. In general, all cost functions are assumed be continuous, non-negative and non-decreasing. Within the time interval $[t_0, t_f]$, consider $N-1$ impulse actions applied at $t = t_n$ for $n = 1, 2, \cdots, N-1$. Then the objective function of this system is as follows:

$$\begin{aligned} J_{OC} =& \mathbb{E}\Bigg\{ \sum_{n=0}^{N-1} \Bigg[\int_{t_n^+}^{t_{n+1}^-} \left(A_1 I_1(t) + A_2 I_2(t) + \frac{1}{2} c_1 u_1^2(t) + \frac{1}{2} c_2 u_2^2(t) \right) \Bigg] dt \\ &+ \sum_{n=1}^{N-1} \left[\frac{1}{2} D_1 \delta_1^2(t_n^-) + \frac{1}{2} D_2 \delta_2^2(t_n^-) + \frac{1}{2} D_3 \delta_3^2(t_n^-) + \frac{1}{2} D_4 \vartheta^2(t_n^-) \right] \Bigg\}, \end{aligned}$$

where, A_i (i=1, 2) serve as the cost weights that minimizes the number of abnormal individuals, the c_i (i=1, 2) are the cost weights linked to the application of continuous control, and D_i (i=1, 2, 3, 4) represent the cost weights for the application of impulse control. The Hamiltonian functions for both continuous and impulsive control are as follows:

$$\begin{aligned} H_c =& A_1 I_1(t) + A_2 I_2(t) + \frac{1}{2} c_1 u_1^2(t) + \frac{1}{2} c_2 u_2^2(t) + \lambda_1(t)\dot{S}(t) + \lambda_2(t)\dot{I}_1(t) + \lambda_3(t)\dot{I}_2(t) \\ &- q_3(t)[\sigma_2 I_2(t) + \sigma_1 I_2(t) M(t)] + \lambda_4(t)\dot{B}(t) + q_4(t)[\sigma_2 I_2(t)] + \lambda_5(t)\dot{M}(t) \\ &+ q_5(t)[\sigma_1 I_2(t) M(t)] + \lambda_6(t)\dot{X}(t) + \lambda_7(t)\dot{Y}(t), \\ H_{d,n} =& \frac{1}{2} D_1 \delta_1^2(t_n^-) + \frac{1}{2} D_2 \delta_2^2(t_n^-) + \frac{1}{2} D_3 \delta_3^2(t_n^-) + \frac{1}{2} D_4 \vartheta^2(t_n^-) + \lambda_1(t_n^+)[S(t_n^+) - S(t_n^-)] \\ &+ \lambda_2(t_n^+)[I_1(t_n^+) - I_1(t_n^-)] + \lambda_3(t_n^+)[I_2(t_n^+) - I_2(t_n^-)] + \lambda_4(t_n^+)[B(t_n^+) - B(t_n^-)] \\ &+ \lambda_5(t_n^+)[M(t_n^+) - M(t_n^-)] + \lambda_6(t_n^+)[X(t_n^+) - X(t_n^-)] + \lambda_7(t_n^+)[Y(t_n^+) - Y(t_n^-)]. \end{aligned}$$

Theorem 3.3 *The optimal control solution of the system* (1)–(2) *is as follows:*

1) *Its continuous and impulsive adjoint equations are as Equations* (14) *and* (15)*, respectively, and transversal condition is as Equation* (16)*;*

2) *The optimal control variables* $u_1(t)$*,* $u_2(t)$*,* $\delta_1(t_n)$*,* $\delta_2(t_n)$*,* $\delta_3(t_n)$*,* $\vartheta(t_n)$ *are as Equation* (17)*.*

$$
\begin{cases}
\frac{d\lambda_1(t)}{dt} =\lambda_1(t)[\alpha_1 I_2(t) + (1-u_1(t))I_2(t) + \omega_1 M(t) + \mu_1 X(t) + b_1] \\
\qquad - \lambda_2(t)\alpha_1 I_2(t) - \lambda_3(t)(1-u_1(t))I_2(t) - \lambda_5(t)(\omega_1 M(t) + \mu_1 X(t)), \\
\frac{d\lambda_2(t)}{dt} = - A_1 + \lambda_2(t)[\theta I_2(t) + \beta_1 + \omega_2 M(t) + \mu_2 X(t) + b_2] - \lambda_3(t)\theta I_2(t) \\
\qquad - \lambda_4(t)\beta_1 - \lambda_5(t)[\omega_2 M(t) + \mu_2 X(t)], \\
\frac{d\lambda_3(t)}{dt} = - A_2 + \lambda_1(t)[\alpha_1 S(t) + (1-u_1(t))S(t)] - \lambda_2(t)[\alpha_1 S(t) - \theta I_1(t)] \\
\qquad - \lambda_3(t)[(1-u_1(t))\mathrm{S}(t) + \theta \mathrm{I}_1(t) - u_2(t) - \omega_3 M(t) - \mu_3 X(t) - b_3] \\
\qquad - \lambda_4(t)u_2(t) - \lambda_5(t)[\omega_3 M(t) + \mu_3 X(t)] + q_3(t)[\sigma_2 + \sigma_1 M(t)] \\
\qquad - q_4(t)\sigma_2 - q_5(t)\sigma_1 M(t), \\
\frac{d\lambda_4(t)}{dt} =\lambda_4(t)b_4, \\
\frac{d\lambda_5(t)}{dt} =\lambda_1(t)\omega_1 S(t) + \lambda_2(t)\omega_2 I_1(t) + \lambda_3(t)\omega_3 I_2(t) - \lambda_5(t)[\omega_1 S(t) + \omega_2 I_1(t) \\
\qquad + \omega_3 I_2(t) - b_5] - \lambda_6(t)\gamma Y(t) + \lambda_7(t)\gamma Y(t) + q_3(t)\sigma_1 I_2(t) - q_5(t)\sigma_1 I_2(t), \\
\frac{d\lambda_6(t)}{dt} =\lambda_1(t)\mu_1 S(t) + \lambda_2(t)\mu_2 I_1(t) + \lambda_3(t)\mu_3 I_2(t) - \lambda_5(t)[\mu_1 S(t) + \mu_2 I_1(t) \\
\qquad + \mu_3 I_2(t)] - \lambda_6(t)[\lambda Y(t) - b_6] + \lambda_7(t)\lambda Y(t), \\
\frac{d\lambda_7(t)}{dt} = - \lambda_6(t)[\lambda X(t) + \gamma M(t)] + \lambda_7(t)[\gamma M(t) + \lambda X(t) + b_7],
\end{cases} \tag{14}
$$

$$
\begin{cases}
\lambda_1(t_n) = \lambda_1\left(t_n^+\right) + \lambda_1\left(t_n^+\right)\left[1 - \delta_1\left(t_n^-\right)\right] + \lambda_4\left(t_n^+\right)\delta_1\left(t_n^-\right), \\
\lambda_2(t_n) = \lambda_2\left(t_n^+\right) + \lambda_2\left(t_n^+\right)\left[1 - \delta_2\left(t_n^-\right)\right] + \lambda_4\left(t_n^+\right)\delta_2\left(t_n^-\right), \\
\lambda_3(t_n) = \lambda_3(t_n^+) + \lambda_3(t_n^+)[1 - \delta_3(t_n^-) - \vartheta(t_n^-)] + \lambda_4(t_n^+)[\delta_3(t_n^-) + \vartheta(t_n^-)], \\
\lambda_4(t_n) = \lambda_4(t_n^+), \\
\lambda_5(t_n) = \lambda_5\left(t_n^+\right), \\
\lambda_6(t_n) = \lambda_6(t_n^+), \\
\lambda_7(t_n) = \lambda_7\left(t_n^+\right),
\end{cases} \tag{15}
$$

$$
\lambda_i(t_f) = 0, \quad i = 1, \cdots, 7, \tag{16}
$$

$$\begin{cases} u_1^*(t) = \min\left\{ \max\left\{ \dfrac{\lambda_3(t)\mathrm{S}(t)I_2(t) - \lambda_1(t)\mathrm{S}(t)I_2(t)}{c_1}, 0 \right\}, 1 \right\}, \\ u_2^*(t) = \min\left\{ \max\left\{ \dfrac{\lambda_3(t)I_2(t) - \lambda_4(t)I_2(t)}{c_2}, 0 \right\}, 1 \right\}, \\ \delta_1^*(t_n^-) = \min\left\{ \max\left\{ \dfrac{\lambda_1\left(t_n^+\right)S\left(t_n^-\right) - \lambda_4\left(t_n^+\right)S\left(t_n^-\right)}{D_1}, 0 \right\}, 1 \right\}, \\ \delta_2^*(t_n^-) = \min\left\{ \max\left\{ \dfrac{\lambda_2(t_n^+)I_1(t_n^-) - \lambda_4(t_n^+)I_1(t_n^-)}{D_2}, 0 \right\}, 1 \right\}, \\ \delta_3^*(t_n^-) = \min\left\{ \max\left\{ \dfrac{\lambda_3\left(t_n^+\right)I_2\left(t_n^-\right) - \lambda_4\left(t_n^+\right)I_2\left(t_n^-\right)}{D_3}, 0 \right\}, 1 \right\}, \\ \vartheta^*(t_n^-) = \min\left\{ \max\left\{ \dfrac{\lambda_3\left(t_n^+\right)I_2\left(t_n^-\right) - \lambda_4\left(t_n^+\right)I_2\left(t_n^-\right)}{D_4}, 0 \right\}, 1 \right\}. \end{cases} \tag{17}$$

Proof The setting of the cost function mentioned above, together with the system (1), inherently satisfies Assumption 3.1. Consequently, the analytical results (10)–(13) established for the general system (3) can be directly applied, yielding the following important conclusions:

$$\begin{aligned} \frac{d\lambda_1(t)}{dt} = -\frac{\partial H_c}{\partial S(t)} =& \lambda_1(t)[\alpha_1 I_2(t) + (1-u_1(t))I_2(t) + \omega_1 M(t) + \mu_1 X(t) + b_1] \\ & - \lambda_2(t)\alpha_1 I_2(t) - \lambda_3(t)(1-u_1(t))I_2(t) - \lambda_5(t)(\omega_1 M(t) + \mu_1 X(t)), \\ \frac{d\lambda_2(t)}{dt} = -\frac{\partial H_c}{\partial I_1(t)} =& -A_1 + \lambda_2(t)[\theta I_2(t) + \beta_1 + \omega_2 M(t) + \mu_2 X(t) + b_2] - \lambda_3(t)\theta I_2(t) \\ & - \lambda_4(t)\beta_1 - \lambda_5(t)[\omega_2 M(t) + \mu_2 X(t)], \\ \frac{d\lambda_3(t)}{dt} = -\frac{\partial H_c}{\partial I_2(t)} =& -A_2 + \lambda_1(t)[\alpha_1 S(t) + (1-u_1(t))S(t)] - \lambda_2(t)[\alpha_1 S(t) - \theta I_1(t)] \\ & - \lambda_3(t)[(1-u_1(t))\mathrm{S}(t) + \theta \mathrm{I}_1(t) - u_2(t) - \omega_3 M(t) - \mu_3 X(t) - b_3] \\ & - \lambda_4(t)u_2(t) - \lambda_5(t)[\omega_3 M(t) + \mu_3 X(t)] + q_3(t)[\sigma_2 + \sigma_1 M(t)] - q_4(t)\sigma_2 \\ & - q_5(t)\sigma_1 M(t), \\ \frac{d\lambda_4(t)}{dt} = -\frac{\partial H_c}{\partial B(t)} =& \lambda_4(t)b_4, \\ \frac{d\lambda_5(t)}{dt} = -\frac{\partial H_c}{\partial M(t)} =& \lambda_1(t)\omega_1 S(t) + \lambda_2(t)\omega_2 I_1(t) + \lambda_3(t)\omega_3 I_2(t) - \lambda_5(t)[\omega_1 S(t) + \omega_2 I_1(t) \\ & + \omega_3 I_2(t) - b_5] - \lambda_6(t)\gamma Y(t) + \lambda_7(t)\gamma Y(t) + q_3(t)\sigma_1 I_2(t) - q_5(t)\sigma_1 I_2(t), \\ \frac{d\lambda_6(t)}{dt} = -\frac{\partial H_c}{\partial X(t)} =& \lambda_1(t)\mu_1 S(t) + \lambda_2(t)\mu_2 I_1(t) + \lambda_3(t)\mu_3 I_2(t) - \lambda_5(t)[\mu_1 S(t) + \mu_2 I_1(t) \\ & + \mu_3 I_2(t)] - \lambda_6(t)[\lambda Y(t) - b_6] + \lambda_7(t)\lambda Y(t), \\ \frac{d\lambda_7(t)}{dt} = -\frac{\partial H_c}{\partial Y(t)} =& -\lambda_6(t)[\lambda X(t) + \gamma M(t)] + \lambda_7(t)[\gamma M(t) + \lambda X(t) + b_7], \\ \lambda_1(t_n) = \lambda_1\left(t_n^+\right) + \frac{\partial H_{d,n}}{\partial S\left(t_n^-\right)} =& \lambda_1(t_n^+) + \lambda_1(t_n^+)[1-\delta_1(t_n^-)] + \lambda_4(t_n^+)\delta_1(t_n^-), \end{aligned}$$

$$\lambda_2(t_n) = \lambda_2\left(t_n^+\right) + \frac{\partial H_{d,n}}{\partial I_1\left(t_n^-\right)} = \lambda_2\left(t_n^+\right) + \lambda_2\left(t_n^+\right)\left[1 - \delta_2\left(t_n^-\right)\right] + \lambda_4\left(t_n^+\right)\delta_2\left(t_n^-\right),$$
$$\lambda_3(t_n) = \lambda_3(t_n^+) + \frac{\partial H_{d,n}}{\partial I_2(t_n^-)} = \lambda_3(t_n^+) + \lambda_3(t_n^+)[1 - \delta_3(t_n^-) - \vartheta(t_n^-)] + \lambda_4(t_n^+)[\delta_3(t_n^-) + \vartheta(t_n^-)],$$
$$\lambda_4(t_n) = \lambda_4\left(t_n^+\right) + \frac{\partial H_{d,n}}{\partial B\left(t_n^-\right)} = \lambda_4\left(t_n^+\right),$$
$$\lambda_5(t_n) = \lambda_5\left(t_n^+\right) + \frac{\partial H_{d,n}}{\partial M\left(t_n^-\right)} = \lambda_5\left(t_n^+\right),$$
$$\lambda_6(t_n) = \lambda_6\left(t_n^+\right) + \frac{\partial H_{d,n}}{\partial X\left(t_n^-\right)} = \lambda_6\left(t_n^+\right),$$
$$\lambda_7(t_n) = \lambda_7(t_n^+) + \frac{\partial H_{d,n}}{\partial Y(t_n^-)} = \lambda_7(t_n^+).$$

Based on the terminal condition equation (7), the following transversal conditions are given as: $\lambda_i(t_f) = 0,\ i = 1, \cdots, 7$.

Then, the optimal conditions are calculated as Equation (18).

$$\begin{cases} \dfrac{\partial H_c}{\partial u_1(t)} = c_1 u_1(t) + \lambda_1(t)\mathrm{S}(t)I_2(t) - \lambda_3(t)\mathrm{S}(t)I_2(t), \\ \dfrac{\partial H_c}{\partial u_2(t)} = c_2 u_2(t) - \lambda_3(t)I_2(t) + \lambda_4(t)I_2(t), \\ \dfrac{\partial H_{d,n}}{\partial \delta_1(t)} = D_1\delta_1(t) - \lambda_1(t_n^+)S(t_n^-) + \lambda_4(t_n^+)S(t_n^-), \\ \dfrac{\partial H_{d,n}}{\partial \delta_2(t)} = D_2\delta_2(t) - \lambda_2(t_n^+)I_1(t_n^-) + \lambda_4(t_n^+)I_1(t_n^-), \\ \dfrac{\partial H_{d,n}}{\partial \delta_3(t)} = D_3\delta_3(t) - \lambda_3(t_n^+)I_2(t_n^-) + \lambda_4(t_n^+)I_2(t_n^-), \\ \dfrac{\partial H_{d,n}}{\partial \vartheta(t)} = D_4\vartheta(t) - \lambda_3(t_n^+)I_2(t_n^-) + \lambda_4(t_n^+)I_2(t_n^-). \end{cases} \tag{18}$$

Taking into account the range of control variables, the solution of Equation (18) is formulated as follows:

$$\begin{cases} u_1^*(t) = \min\left\{\max\left\{\dfrac{\lambda_3(t)S(t)I_2(t) - \lambda_1(t)S(t)I_2(t)}{c_1}, 0\right\}, 1\right\}, \\ u_2^*(t) = \min\left\{\max\left\{\dfrac{\lambda_3(t)I_2(t) - \lambda_4(t)I_2(t)}{c_2}, 0\right\}, 1\right\}, \\ \delta_1^*(t_n^-) = \min\left\{\max\left\{\dfrac{\lambda_1\left(t_n^+\right)\mathrm{S}\left(t_n^-\right) - \lambda_4\left(t_n^+\right)\mathrm{S}\left(t_n^-\right)}{D_1}, 0\right\}, 1\right\}, \\ \delta_2^*(t_n^-) = \min\left\{\max\left\{\dfrac{\lambda_2(t_n^+)I_1(t_n^-) - \lambda_4(t_n^+)I_1(t_n^-)}{D_2}, 0\right\}, 1\right\}, \\ \delta_3^*(t_n^-) = \min\left\{\max\left\{\dfrac{\lambda_3\left(t_n^+\right)I_2\left(t_n^-\right) - \lambda_4\left(t_n^+\right)I_2\left(t_n^-\right)}{D_3}, 0\right\}, 1\right\}, \\ \vartheta^*(t_n^-) = \min\left\{\max\left\{\dfrac{\lambda_3\left(t_n^+\right)I_2\left(t_n^-\right) - \lambda_4\left(t_n^+\right)I_2\left(t_n^-\right)}{D_4}, 0\right\}, 1\right\}. \end{cases}$$

Therefore, Theorem 3.3 has been proven. ∎

3.3 Data-Driven Optimization Control Algorithm

In order to enhance the adaptability and make up for the repeated iterative calculation problem of optimal control (OC) method, we introduce data-driven optimization control (DOC) algorithm, which consists of three parts: Data collection, training process and controlling, the specific process is shown in Figure 4.

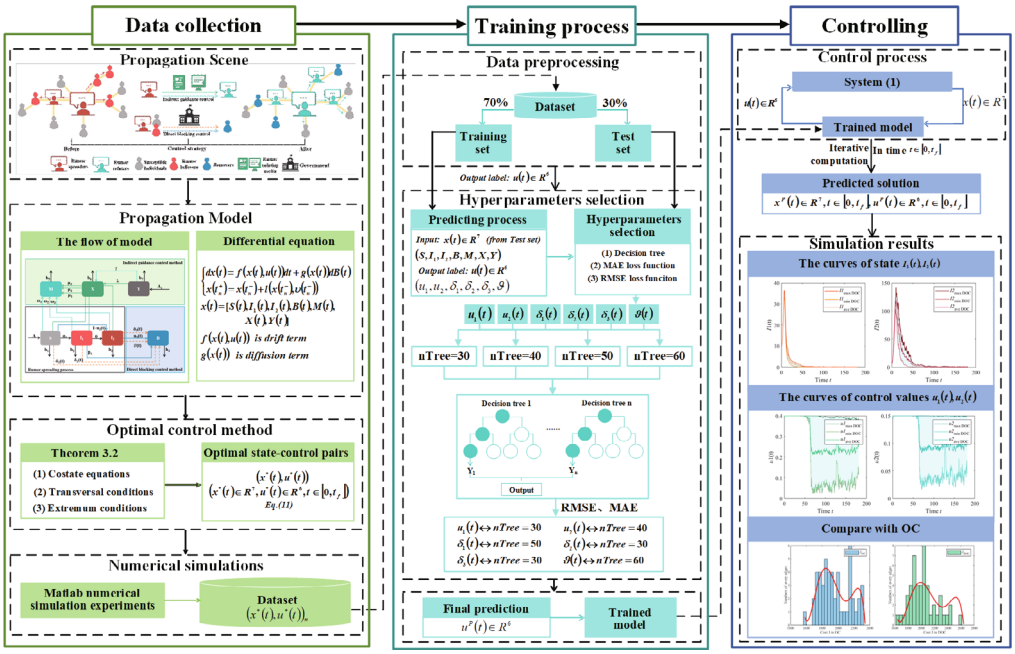


Figure 4 Flow of DOC algorithm

3.3.1 Data Collection

The process of data collection is depicted in the leftmost rectangle of Figure 4. Firstly, the optimal control pair $(x^*(t),\ u^*(t))$, $x^*(t) \in R^7$, $u^*(t) \in R^6$, $t \in [0, t_f]$ of the original model was obtained through the calculation of OC method, and it was used as the dataset for DOC algorithm that will be applied. $x^*(t)$, $u^*(t)$ are defined in Equation (19). Taking into account the different initial states of the system within a specified range, more samples can be effortlessly acquired, resulting in the generation of 1,086,000 samples from 600 distinct initial state groups. Here, $x^*(t)$ are regarded as input label and $u^*(t)$ are regarded as output label. It should be highlighted that the OC state pair $u^*(t)$ can be solved by the OC method (Theorem 3.3), and the introduced DOC algorithm is capable of dealing with stochastic or deterministic hybrid OC problems.

$$\begin{cases} x^*(t) = [S^*(t), I_1^*(t), I_2^*(t), B^*(t), M^*(t), X^*(t), Y^*(t)]\,, \\ u^*(t) = [u_1{}^*(t), u_2{}^*(t), \delta_1{}^*(t), \delta_2{}^*(t), \delta_3{}^*(t), \vartheta^*(t)]\,. \end{cases} \tag{19}$$

3.3.2 Training Process

The training process is depicted in the central rectangle of Figure 4. This process is divided into three stages: Data preprocessing stage, hyperparameter selection stage and training stage. In the data preprocessing phase, 70% of the datasets were allocated to training set and the other data were allocated to test set. Initially, the training set is used for prediction, and the input comes from the test set. In this stage, we employed the Random Forest algorithm, an ensemble learning algorithm known for its high accuracy and broad applicability in solving regression problems, for model training. Considering the diversity and complexity of decision tree models, we selected the number of decision trees as the hyperparameter for each control variable $u^*(t) \in R^6$, $t \in [0, t_f]$, setting four groups of hyperparameters (30, 40, 50, and 60) for the six control variables. This setup allowed us to construct different decision tree models. Using the state variables $x^*(t) \in R^7(S, I_1, I_2, B, M, X, Y)$ from the training set as input labels and the control variables $u^*(t) \in R^6(u_1, u_2, \delta_1, \delta_2, \delta_3, \vartheta)$ from the control group as output labels, we trained and predicted results using the Random Forest algorithm for the aforementioned models. We evaluated and compared the model performance based on the prediction results, primarily using the mean absolute error (MAE) and root-mean-square error (RMSE) between the predicted control variables $u^*(t) \in R^6(u_1, u_2, \delta_1, \delta_2, \delta_3, \vartheta)$ and those in the test set. The specific calculation results are shown in Table 2. Subsequently, for each control variable, we selected the model with the best performance evaluation results, i.e., the smallest MAE and RMSE values, as the final trained model. In the training phase, we imported the best-performing decision tree models for training to obtain the final trained model.

Table 2 Hyperparameter selection

Parameter	Hyper-parameter	MAE	RMSE	Parameter	Hyper-parameter	MAE	RMSE
	30	**0.0536775**	**0.0803497**		30	0.0086017	0.0176410
u_1	40	0.0538381	0.0808538	u_2	**40**	**0.0083221**	**0.0169313**
	50	0.0538732	0.0804171		50	0.0084481	0.0173980
	60	0.0543815	0.0811505		60	0.0083819	0.0171297
	30	0.0001093	0.0016542		**30**	**3.41E-07**	**4.89E-06**
δ_1	40	0.0001103	0.0015939	δ_2	40	5.25E-07	6.73E-06
	50	**0.0001101**	**0.0015339**		50	4.82E-07	6.34E-06
	60	0.0001075	0.0015889		60	4.01E-07	5.44E-06
	30	**8.28E-05**	**0.0014045**		30	8.67E-05	0.0015346
δ_3	40	8.45E-05	0.0015398	ϑ	40	9.78E-05	0.0015734
	50	8.82E-05	0.0014810		50	8.57E-05	0.0014556
	60	9.36E-05	0.0014955		**60**	**9.20E-05**	**0.0013861**

3.3.3 Controlling

The process of controlling is depicted in the rightmost rectangle in Figure 4. Instead of using the *OC* method, we use the trained model to predict the control solution of the system. Given an initial value for the newly trained model, that is, selecting an initial sample $x^*(0)$ from the test set as the initial input label for the well-trained model, and providing the model with parameters

related to the propagation environment, the output label, which is a pair of control variables, is obtained through the prediction of the trained model. This predicted pair of control variables is then applied to the rumor propagation control system we have established, generating a new input label $x^P(1)$, which will serve as a new input label for the well-trained model. Thus, through such iterative predictions, the complete state variable trajectory $x^P(t), t \in [0, t_f]$ and control variable trajectory $u^P(t), t \in [0, t_f]$ are obtained. Ultimately, the obtained states $x^*(t) \in R^7(S, I_1, I_2, B, M, X, Y)$ and control $u^*(t) \in R^6(u_1, u_2, \delta_1, \delta_2, \delta_3, \vartheta)$ are presented in the form of state variable trend charts and control variable change curve charts, and the control cost statistical chart is obtained through the calculation of the cost function.

4 Numerical Simulation

In this section, based on real cases data, we design simulation experiments to verify the rationality and accuracy of the model (1) (Subsection 4.1); Validity of OC theory (Subsection 4.2); Effectiveness and applicability of the ensemble learning control algorithm (Subsection 4.3). The simulation results show that our newly established the model (1) can well simulate the real case propagation trend, and the inherited learning control algorithm can intelligently find the rumor OC scheme.

4.1 Propagation and Real Data Fitting

To verify the accuracy of the model (1), we collected two real rumor cases data as support to fit an accurate and feasible model. The accuracy of the model (1) was ensured by fitting the dynamic relationship between the propagation curve of individuals I_2 and the propagation in real cases.

Among the two real rumor cases collected, the first event is "Rumor Event of China Auxiliary Police Will Be Fully Regular", and the relevant data of the event comes from the Zhiwei data platform. On January 26th, 2024 from 9:00 to 18:00 for data sampling, the sampling frequency is three times an hour (data source: https://ef.zhiweidata.com/), the final as Figure 5(a). The scatter points in the figure are used to represent the overall changing trend of the real case data curve. Model fitting parameters by least squares method, and the parameters are listed in the "Original propagation control system" section of Table 3, and the initialization values are listed in the "Initial states value" section of Table 3. The obtained parameters were substituted into the numerical simulation experiment to obtain the fitted model data, as shown by the curve in Figure 5(a).

The model (1) can also predict other rumor transmission events. The second event is "80%, Tsinghua University Graduate go abroad Rumors" in the data (data source: https://ef.zhiweidata.com/), on January 17th, 2024 from 7:00 to 14:00 for data sampling, the sampling frequency is three times an hour, and the final result is Figure 5(b). The scatter in Figure 5(b) shows the overall change trend of the real data curve, and the curve represents the change curve of the model (1).

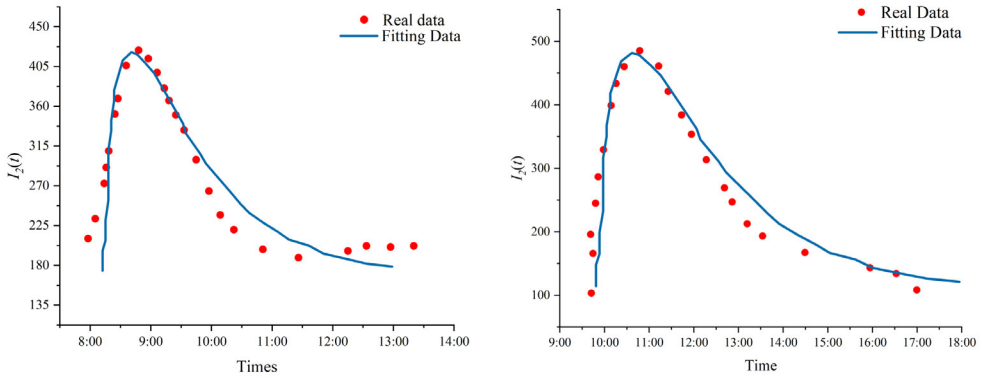


Figure 5 The line of $I_2(t)$ and scatter diagram of real case data. (a) "80% of Tsinghua University Graduates go aboard Rumors"; (b) "Auxiliary Police Will Be Fully Regular Rumors"

Table 3 The values of relevant parameters

Type	Parameter	Value	Parameter	Value	Parameter	Value	Parameter	Value
	A	100	A_z	20	b_1	0.2	b_2	0.2
Original	b_3	0.2	b_4	0.2	b_5	0.2	b_6	0.2
propagation	b_7	0.2	α_1	0.001	α_2	0.003	β_1	0.02
control	θ	0.001	λ	0.001	γ	0.001	ω_1	0.001
system	ω_2	0.0091	ω_3	0.0001	μ_1	0.0001	μ_2	0.0009
	μ_3	0.0009	σ_1	0.0005	σ_2	0.005	q_3	0.01
	q_4	0.01	q_5	0.01				
Cost	A_1	1	A_2	1	c_1	2	c_2	2
coefficient	D_1	2	D_2	2	D_3	2	D_4	2
Initial	S	480	I_1	10	I_2	5	B	3
states value	M	2	X	5	Y	95		

Figure 5 shows the dynamic relationship between the individual I_2 in the fitted curve and the actual situation curve. Through comparative observation, it can be seen that the scatter and curve in Figure 5(a) and Figure 5(b) can fit well, indicating that the model (1) can have a good fitting result with the rumor propagation in the real case, and also indicating that the model (1) can well predict other rumor propagation events. Therefore, the accuracy of the models (1)–(2) can be verified by simulation experiments.

4.2 Simulation Results of the OC Theory

In this subsection, we verify the effectiveness of the OC strategies, namely Theorem 3.3, based on real cases data. Firstly, it is not difficult to observe Figure 5 that without any control scheme, the rumor will continue to spread. According to Theorem 3.3, we adopt the $DR-DB$ control strategy and design the cost function (20) and the OC strength as the formula (17). Then, we use the Matlab simulation tool to select the system parameters fitted by the real cases

in Table 3 and design the iterative algorithm to carry out 600 rounds simulation experiments, and the final results are shown in Figure 6. The first and second rows of Figure 6 show the trend curves of the state variables of the propagation nodes, the continuous control variables, and the impulse control variables. Comparing with Figure 5, it can be seen that the rumor spreaders almost certainly tends to zero, and the rumor is effectively suppressed. It should be particularly pointed out that our control strategy is stochastic, so we carried out 600 rounds of simulation and presented all the simulation results, obtaining the charts shown in Figure 6.

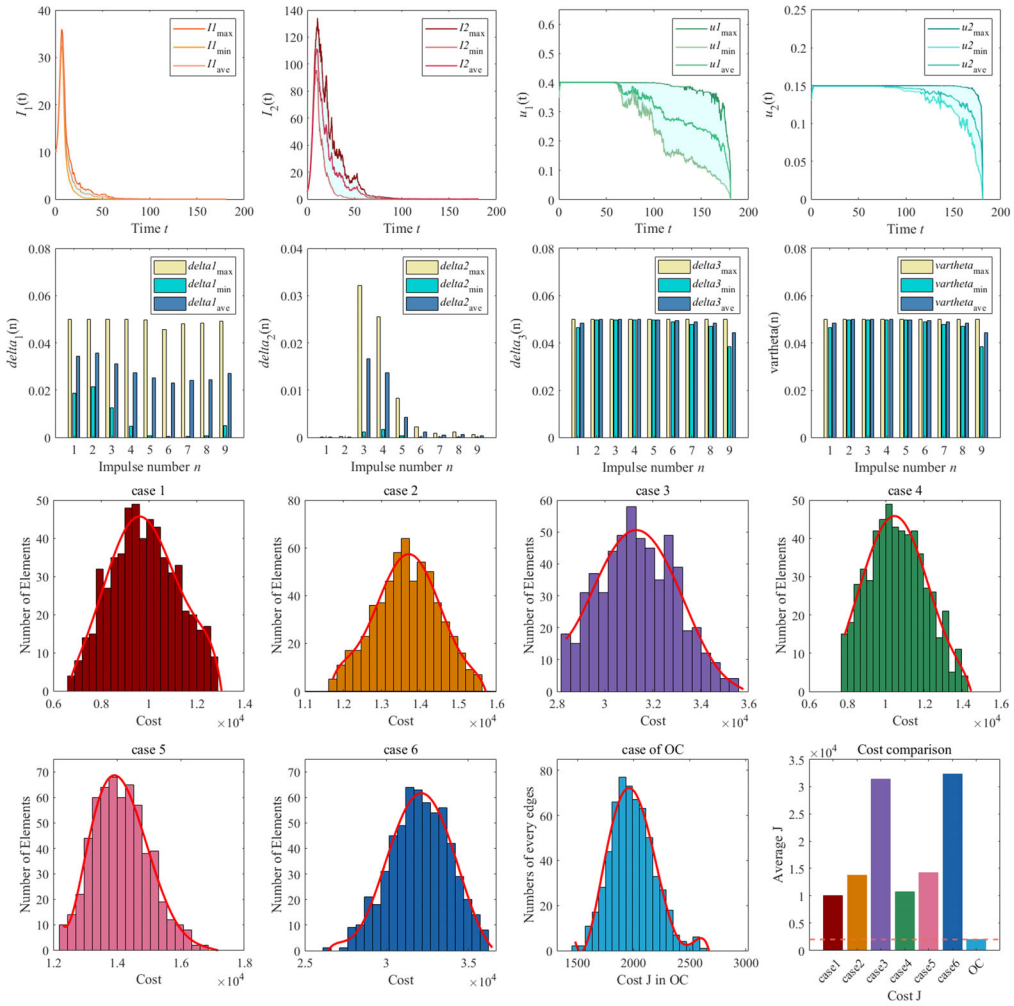


Figure 6 The *OC* theory simulation experiment results. The initial two rows show the trend curves for the state variables $I_1(t)$, $I_2(t)$, the continuous control variables $u_1(t)$, $u_2(t)$ and the impulse control variables $\delta_1(t)$, $\delta_2(t)$, $\delta_3(t)$, $\vartheta(t)$. The latter two rows are dedicated to showcasing special cases, including the cost statistical charts of six unique scenarios and the *OC* theory. The final chart in the sequence presents the average cost across different control strategies

We select a cost function with

$$J_{OC} = \mathbb{E}\Bigg\{ \sum_{n=0}^{N-1} \Bigg[\int_{t_n^+}^{t_{n+1}^-} \left(A_1 I_1(t) + A_2 I_2(t) + \frac{1}{2} c_1 u_1^2(t) + \frac{1}{2} c_2 u_2^2(t) \right) \Bigg] dt + \sum_{n=1}^{N-1} \left[\frac{1}{2} D_1 \delta_1^2(t_n^-) + \frac{1}{2} D_2 \delta_2^2(t_n^-) + \frac{1}{2} D_3 \delta_3^2(t_n^-) + \frac{1}{2} D_4 \vartheta^2(t_n^-) \right] \Bigg\}. \tag{20}$$

To verify the accuracy of the OC strategy, we also select six special types of control strategies, including:

• Comprehensive dynamic rumor propagation punishment control (Case 1), its total cost increases by 80.44%.

• Comprehensive dynamic rumor continuous isolation control (Case 2), its total cost increases by 85.72%.

• Comprehensive fixed intensity rumor control (Case 3), its total cost increases by 93.73%.

• Local dynamic continuous control (Case 4), its total cost increases by 81.59%.

• Local dynamic blocking control (Case 5), its total cost increases by 86.16%.

• Local fixed intensity control (Case 6), its total cost increases by 93.91%.

The third and fourth rows of Figure 6 show the cost statistical graph under different special control schemes, and the far right graph in the fourth row of Figure 6 shows the average cost comparison of all control strategies. Among them, the OC strategy has the least cost, and simulation experiments verify the effectiveness of the OC strategy.

4.3 Simulation Results of DOC Algorithm

In this subsection, our aim is to further prove that under the premise of ensuring the OC effect, our proposed DOC algorithm can improve the adaptability of OC theory through simulation experiments. According to the control strategy flow designed in Figure 4, the simulation results of DOC algorithm numerical simulation using the dataset obtained by OC theory simulation are shown in Figure 7. Comparing the first and second rows of Figure 6 and Figure 7, namely the state and control variable curves under OC theory and DOC algorithm, it can be seen that DOC algorithm can also find the OC strategy. More specifically, we compare the state variable curves under OC theory and DOC algorithm, as shown in the left charts 1, 2 in the third row of Figure 7, which almost remain the same.

From the point of view of controlling cost, we similarly make a specific comparison. The two statistical graphs in the third row of Figure 7 show the rumor control cost under OC theory and DOC algorithm. The specific comparative data between J_{OC} and J_{DOC} are shown in Table 4. Table 4 lists the cost results calculated by both OC theory and DOC algorithm, comparing their means, MAE, and $RMSE$ metrics. The comparative data reveal that the cost of OC theory is smaller than that of DOC algorithm, and its MAE and $RMSE$ metrics outperform those of DOC algorithm, which is consistent with the actual results of optimal control. However, the results of DOC algorithm are very close to those of OC theory, with the comparison indicating that the control cost of DOC algorithm is only 4.1% higher than that of OC theory. The data demonstrate that DOC algorithm can replace the computational approach of OC theory,

obtaining highly consistent and nearly identical prediction results through a fast and efficient method, thereby addressing the issues of complex iterative calculations and high computational difficulty associated with *OC* theory. Moreover, compared to *OC* theory, *DOC* algorithm is more adaptable to initial values and system parameters, and can effectively obtain *OC* strategy.

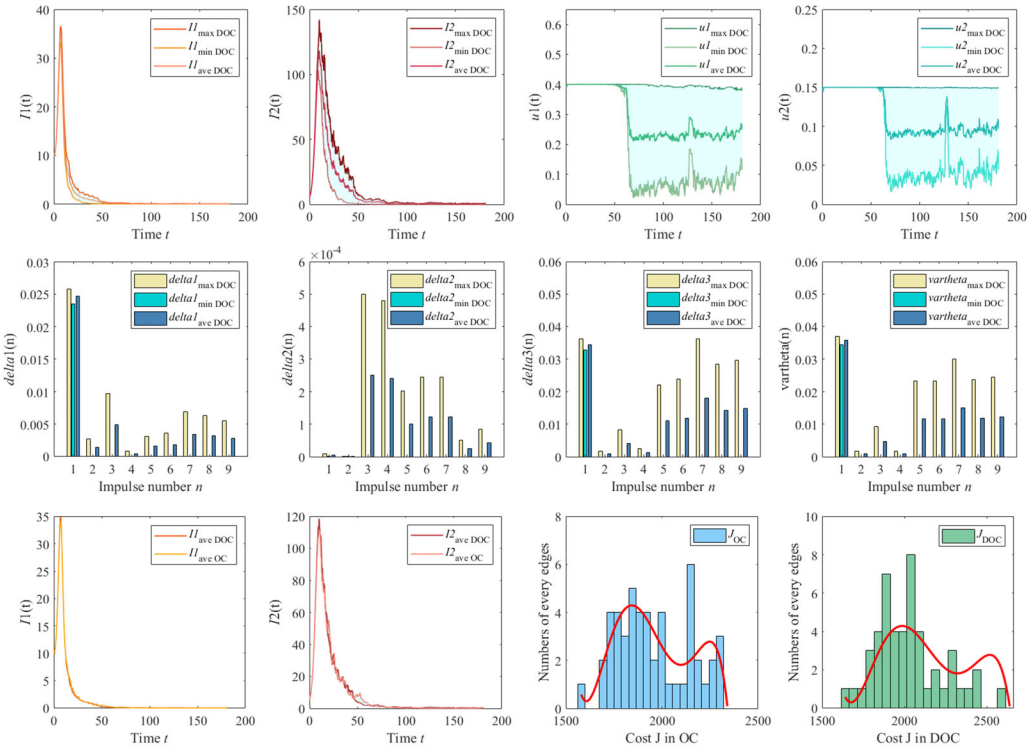


Figure 7 The *DOC* algorithm simulation experiment results. The initial two rows show the trend curves for the state variables $I_1(t)$, $I_2(t)$, the continuous control variables $u_1(t)$, $u_2(t)$ and the impulse control variables $\delta_1(t)$, $\delta_2(t)$, $\delta_3(t)$, $\vartheta(t)$. The latter row is the average results of $I_1(t)$ and $I_2(t)$ and the cost statistical charts under the *OC* theory and *DOC* algorithm respectively

Table 4 The comparison of the statistical characteristics between J_{OC} and J_{DOC}

Parameter	J_{OC}	J_{DOC}
Average	1976.076905	2060.94502
MAE	161.199566	167.3234984
RMSE	188.614997	215.4989938

Overall, *DOC* algorithm we designed is able to quickly identify the sub-optimal control strategy for rumor propagation, with costs only 4.1% higher than those of optimal control theory. In future research, we will continue to explore the scalable theoretical foundations underlying the algorithm.

4.4 Practical Application Analysis

The $DR-DB$ control model (1)–(2) provides a mathematical description of the real rumor control process and can be used to analyze and predict the trend of rumor propagation events, offering trend curves for the entire rumor propagation chain. Figure 5 shows the real data and model simulation curves for two rumor propagation cases. A comparative analysis reveals that the model curves can well fit the trend of real data changes. Therefore, given data restructuring, the $DR-DB$ control model can analyze and predict rumor spread trends, providing quantitative evaluation indicators for government or related departments.

The comparative analysis in Subsection 4.2 demonstrates that the coordinated stochastic hybrid rumor control strategy, combining direct blocking and guided rumor refutation, significantly reduces rumor control costs compared to single control strategies while effectively controlling rumor propagation. In real rumor control scenarios, fitting propagation parameters based on existing propagation data allow for preliminary acquisition of relevant intervention intensity and the strength of accurate information dissemination at different time nodes according to Theorem 3.3. By training the control algorithm with existing rumor control data, a final data-driven optimization control algorithm can be developed to address rumor control issues under model parameter changes, adaptively providing quantitative intervention intensity and accuracy in information dissemination.

5 Conclusion

In this article, we designed double refutation (DR) and double blocking (DB) control strategies, and establish a dual-layer propagation control model. From the perspective of minimizing the control cost, we fomulated an OC problem and obtained the necessary conditions for an optimal system by the variational methods. In particular, an ensemble learning algorithm has been designed to substitute the theoretical derivation process, which improves the response speed of the control strategy and the lag defect of the control strategy. Numerical simulation shows that the trained ensemble learning control algorithm can quickly find the sub-OC strategy of rumor spreading, with costs that are only 4.1% higher than those of OC theory.

The synergy of DR and DB control strategies can improve the success rate of dispelling rumors and reduce the control cost. Based on the aforementioned theories and simulation results, guidance can be provided for controlling rumors in the real world. However, the propagation of rumors is a complex process, closely associated with user psychology, group interactions, and the random propagation environment. If the network topology can be improved by further considering hyper relationships among groups, and by describing the time-varying transmission rates through psychological and environmental factors, the model might more closely align with actual propagation scenarios. Furthermore, in the proposed DOC algorithm, consideration could also be given to incorporating algorithms such as reinforcement learning to reduce dependency on data and enhance the applicability of the control algorithm. The aforementioned issues will serve to significantly advance the research in rumor prediction and control, which will also be the focus of our subsequent investigative efforts.

Conflict of Interest

The authors declare no conflict of interest.

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